

● EASY

● GEOMETRY AND TRIGONOMETRY

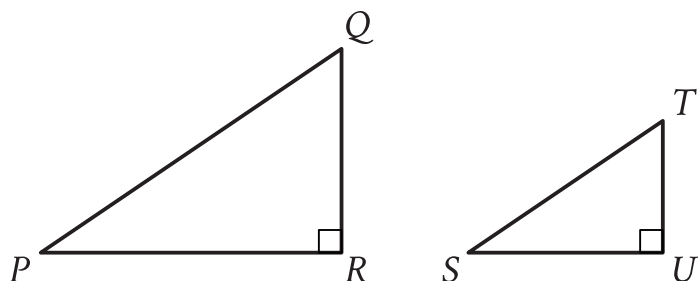
Lines, angles, and triangles

5 questions · ~8 min

04

Q1

GEOMETRY AND TRIGONOMETRY

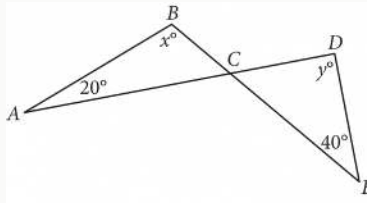


Note: Figures not drawn to scale.

- Right triangle PQR is labeled as follows:
 - Angle Q is a right angle.
- Right triangle STU is labeled as follows:
 - Angle U is a right angle.
- A note indicates the figures are not drawn to scale.

Right triangles PQR and STU are similar, where P corresponds to S . If the measure of angle Q is 18° , what is the measure of angle S ?

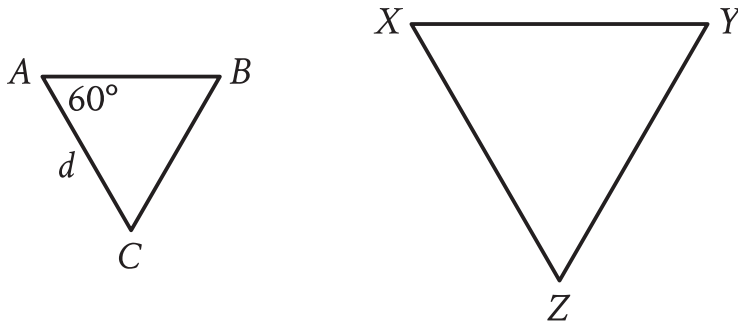
- A. 18°
- B. 72°
- C. 82°
- D. 162°



Note: Figure not drawn to scale.

In the figure above, $\overline{AD}$ intersects $\overline{BE}$ at C. If $x = 100$, what is the value of y ?

- A. 100
- B. 90
- C. 80
- D. 60



Note: Figures not drawn to scale.

- The 2 triangles are oriented such that upper A corresponds to upper X, upper B corresponds to upper Y, and upper C corresponds to upper Z.
- Triangle upper A upper B upper C is labeled as follows:
 - The measure of angle upper A is 60° .
 - The length of side upper A upper C is d .
- A note indicates the figures are not drawn to scale.

For the triangles shown, triangle ABC is dilated by a scale factor of 3 to obtain triangle XYZ , where $d = 16$. What is the measure, in degrees, of angle X ?

- A. 20
- B. 57
- C. 60
- D. 63

In triangle ABC , the measure of angle B is 52° and the measure of angle C is 17° . What is the measure of angle A ?

- A. 21°
- B. 35°
- C. 69°
- D. 111°

In $\triangle XYZ$, the measure of $\angle X$ is 23° and the measure of $\angle Y$ is 66° . What is the measure of $\angle Z$?

- A. 43°
- B. 89°
- C. 91°
- D. 179°